# 12.4 Linear functions

**Before calc 3:** If a line goes through $(x_0, y_0)$ and has slope $m$, it has equation

$$y - y_0 = m(x - x_0)$$

**Calc 3 version:** If a plane goes through $(x_0, y_0, z_0)$ and has slope $m$ in the $x$-direction and slope $n$ in the $y$-direction, it has equation

$$z - z_0 = m(x - x_0) + n(y - y_0). \quad \leftarrow \text{"point-slope"}$$

A **linear function** is a function $z = f(x,y)$ that looks like

slope in x-direction

slope in y-direction

$$z = mx + ny + c \quad \leftarrow \text{"slope-intercept"}$$

z-intercept

→ Do problems 1-3.